

Smart Public Transport Analytics & Business Intelligence Dashboard

Analytics Report

Teyzix Core - Data Analytics Internship - Task DA-3

1. Executive Summary

This report presents an end-to-end business intelligence solution for a public transport operation, developed as part of the Teyzix Core Data Analytics Internship (Task DA-3). The project covers the full analytics lifecycle: creating an original dataset, cleaning and validating it, performing exploratory analysis, developing key performance indicators, and presenting the findings through an interactive Power BI dashboard.

Across four months of operation, the network ran 1,300 trips over eight routes, carrying **42,025 passengers and generating Rs3.75 million in ticket revenue**. Average delay was 11.1 minutes, average occupancy 73.4%, and 91.7% of scheduled trips were completed. The analysis identified three headline findings: weather is the primary driver of delays, ridership is heavily concentrated in morning and evening commuter peaks, and the Electric fleet is dramatically more fuel-efficient than other bus types.

2. Data Cleaning Summary

An original dataset of 1,335 raw records (including deliberate data-quality issues) was cleaned and validated down to 1,300 reliable records. The following issues were identified and resolved:

Issue	Resolution
Duplicate records (35)	Removed with de-duplication
Missing values (passengers, weather, fuel, driver)	Numeric filled with median (fuel by bus type); categorical with mode / "Unknown"
Inconsistent labels (bus type, weather, day type)	Standardised 15 bus-type and 13 weather variants to clean categories
Mixed date & time formats	Parsed to a single standard date and 24-hour time format
Text-encoded numbers ("Rs 2,200", "6.9 L")	Stripped symbols and converted to numeric types
Validation errors (arrival before departure, negative delay)	Reversed times corrected; impossible values treated as missing
Outliers (over-capacity, absurd delays, extreme fuel)	Capped or removed using capacity limits and IQR thresholds

The final dataset contains zero missing values, zero duplicates, standardised categories, correct data types, and no logically impossible records.

3. Key Performance Indicators

Nine core KPIs were developed to monitor the operation:

KPI	Value
Total Passengers	42,025
Total Revenue	Rs3.75M
Average Daily Passengers	350
Average Delay Time	11.1 minutes
Average Occupancy Rate	73.4%
Trip Completion Rate	91.7%
Most Profitable Route	R01 – City Center – Airport (Rs725,496)
Least Performing Route	R06 – Old Town – Hospital (Rs287,548)
Peak Travel Hour	7:00 AM

4. Dashboard

An interactive Power BI dashboard was developed to present the analysis. It features an eight-KPI executive summary strip, seven distinct visual types (bar, line, area, column, histogram, donut, and scatter charts), and four interactive slicers (Route, Trip Status, Weather, and Bus Type) that filter every visual simultaneously.

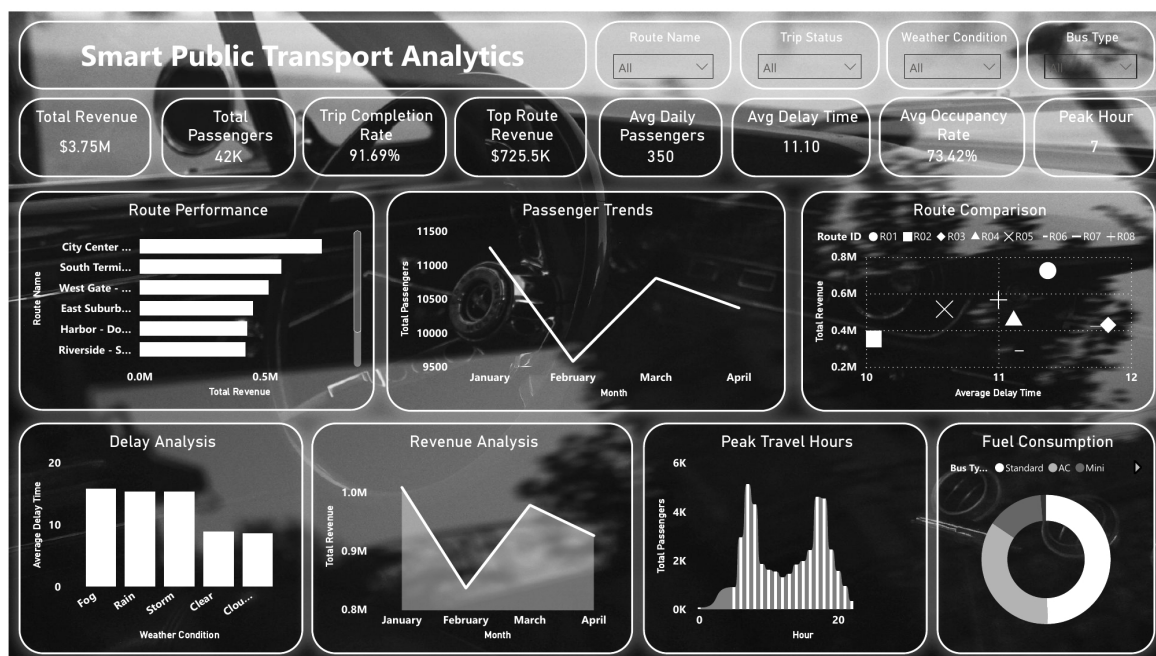


Figure 1: Smart Public Transport Analytics dashboard (Power BI).

5. Business Insights

- **Highest revenue routes are fare-driven.** R01 (Airport) leads at Rs725K due to premium fares, not passenger volume.
- **Weather is the primary delay driver.** Delays nearly double in fog, rain, and storms (≈ 15 min) versus clear weather (≈ 9 min).
- **Demand peaks sharply at commuter hours.** 7–8 AM and 5–6 PM dominate ridership; 7 AM is busiest at 5,140 passengers.
- **Electric buses are far more efficient.** They use ~ 0.03 L per passenger versus 0.57 for Mini — roughly a tenfold difference.
- **Cancellations concentrate on specific routes.** 8.3% of trips are cancelled overall, peaking on R05 (10.8%) and R08 (9.3%).

6. Recommendations

1. **Weather-aware scheduling** — add buffer times during poor weather to keep timetables reliable.
2. **Peak-hour fleet planning** — concentrate high-capacity buses at 7–8 AM and 5–6 PM; trim low-demand mid-day and weekend service.
3. **Expand the Electric fleet** — replace Mini and diesel buses to cut fuel cost and emissions.
4. **Optimise routes** — review low-revenue routes (R06, R02) and protect high-fare corridors (R01).
5. **Reduce cancellations** — improve vehicle and driver scheduling on R05 and R08 to lift service reliability.

7. Conclusion

This project delivered a complete business intelligence solution, transforming a raw, self-created transport dataset into a clean, validated foundation for analysis and an interactive decision-support dashboard. The operation is stable and well-utilised; its clearest opportunities are operational — managing weather-driven delays, planning fleet capacity around predictable commuter peaks, expanding the efficient Electric fleet, and reducing cancellations. Acting on these data-driven recommendations would improve reliability, lower operating costs, and enhance the passenger experience.